

Manufacturer	Customer	Business Transaction	
Example Forge Works GmbH Stahlweg 8 47051 Duisburg DE info@example-forge.com	Example Steel Trading GmbH Musterstrasse 10 70173 Stuttgart DE info@example-customer.com	Order	ORD-2025-3312 · 18000 kg
		Delivery	DLV-2025-4455 · 18000 kg

Product

Product Name	Forged round bars	Name (EN 10027-1)	S355J2 +N
Batch ID	BATCH-2025-03-7832	Number (EN 10027-2)	1.0577 +N
Surface Condition	Rough turned	Shape	RoundBar - D 350 mm - Length 2800 mm
Delivery Condition	+N - (Normalized)	Marking	Both ends stamped with grade and heat number.
		Packaging	Bundled, max 20 MT
		Coloring	blue
		Product Norm	EN 10250-1
		Product Norm	EN 10250-2

Heat Treatment

Process: Normalizing - Lot: HT-LOT-2025-0315-B - Furnace: FURNACE-07 - Date: 2025-03-15				
Stage	Temperature	Duration	Cooling	Atmosphere
Austenitizing	910 C			Air

Chemical Analysis

Heat Number H-2025-7832 - Melting Process EAF+LF+VD - Casting Method IngotCasting																				
Symbol	C	Si	Mn	P	S	Cr	Ni	Mo	Cu	B	Al	V	Sn	Ti	Sb	Ti	Sb	N	Ca	Nb
Unit	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	%	%	%	%
Min	0.10		1.10		0.015						0.015									
Max	0.22	0.55	1.65	0.030	0.030	0.35	0.35	0.10	0.30		0.060	0.025	0.015	0.006	0.0030	3.0	0.50	0.012	0.0030	0.030
Actual	0.17	0.28	1.38	0.014	0.022	0.18	0.14	0.05	0.22	0.0004	0.028	0.012	0.006	0.004	0.0012	2.1	0.38	0.009	0.0018	0.015
Symbol	Co	W	F1																	
Unit	%	%	%																	
Min																				
Max	0.15	0.05	0.52																	
Actual	0.08	0.02	0.37																	

F1 = Cr+Ni+Mo: 0.37%

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	
Tensile Strength¹ Specimen: 1/2T, L - ID: TS-001	Rm	548 N/mm ²	440 N/mm ²	630 N/mm ²	EN ISO 6892-1/B1	
Yield Strength¹ Specimen: 1/2T, L - ID: YS-001	Re	372 N/mm ²	320 N/mm ²		EN ISO 6892-1	
Elongation¹ Specimen: 1/2T, L - ID: EL-L-001	A	26 %	17 %		EN ISO 6892-1/B1	
Elongation¹ Specimen: 1/2T, T - ID: EL-T-001	A	20 %	14 %		EN ISO 6892-1/B1	
Reduction of area¹ Specimen: 1/2T, L - ID: RA-001	Z	48 %			EN ISO 6892-1	
Impact test ISO-V² Specimen: 1/2T, T - ID: IT-T-001					EN ISO 148-1	
No.	1	2	3	Average	Min	Spec Min
Value [J]	36	30	37	34.33	22	22
Impact test ISO-V² Specimen: 1/2T, L - ID: IT-L-001						EN ISO 148-1
No.	1	2	3	Average	Min	Spec Min
Value [J]	62	65	58	61.67	38	38
Surface hardness		162 HB		180 HB		

¹ in delivery condition ² -20°C, in delivery condition

Physical Properties

Property	Actual	Target/Min	Maximum	Method
Grain size	6	4	9	ASTM E112
Purity K4(ox)	20		35	DIN 50602
Purity K4(s)	25		35	DIN 50602
Straightness	1.2 mm/m		2.0 mm/m	
Reduction ratio	3.8	3.0		
Residual magnetism	5 A/cm		10 A/cm	

✓ fully formed structure, no residue of dendritic cast structure

Supplementary Tests

Test	Result / limits	Method
Ultrasonic test 100%	Yes - Pass	EN 10228-3 class 3
Radioactivity	Yes - max 100 Bq/kg	

✓ 100% visually inspected, surface free from defects

Validation

We hereby certify that the material described above has been manufactured, tested and inspected in accordance with the requirements of the applicable specifications and standards, and that the results comply with the specified requirements.
✓ Material is of non-Russian origin (EU Regulation No. 833/2014)

Validated by **Dr. Maria Fischer**, Head of Quality Assurance, Quality Assurance - 2025-04-10